

Milestone 1
DB Project, Winter 2025
Relational Model Solution

UNIVERSITY (**University_ID**, University_Name, Acronym, Established_Year, Accreditation_Body, Address, Contact_Email, Website_URL)

FACULTY (**Faculty_ID**, Faculty_Name, Location, Contact_Email, University_ID)

- References:
 - University_ID → UNIVERSITY.University_ID

DEPARTMENT (**Department_ID**, Department_Name, Department_Type, Location, Contact_Email)

ACADEMIC_DEPARTMENT(**Department_ID**, Faculty_ID)

- References:
 - Department_ID → DEPARTMENT.Department_ID
 - Faculty_ID → FACULTY.Faculty_ID

ADMINISTRATIVE_DEPARTMENT(**Department_ID**, University_ID)

- References:
 - Department_ID → DEPARTMENT.Department_ID
 - University_ID → UNIVERSITY.University_ID

EMPLOYEE (**Employee_ID**, First_Name, Middle_Name, Last_Name, Arabic_Name, Gender, Nationality, DOB, Place_of_Birth, Marital_Status, Religion, Employment_Status, Mobile_Phone, Work_Phone, Work_Email, Personal_Email, Emergency_Contact_Name, Emergency_Contact_Phone, Emergency_Contact_Relationship, Residential_City, Residential_Area, Residential_Street, Residential_Country, Permanent_City, Permanent_Area, Permanent_Street, Permanent_Country, Medical_Clearance_Status, Criminal_Status)

EMPLOYEE_DISABILITY (**Disability_ID**, Employee_ID, Disability_Type, Severity_Level, Required_Support)

- References:
 - Employee_ID → EMPLOYEE

Note, EMPLOYEE_DISABILITY can be also treated as weak entity. However, for audit, consistency and ease of implementation, use the above design.

EMPLOYEE_DISABILITY (**Employee_ID**, **Disability_Type**, Severity_Level, Required_Support)

- References: **Employee_ID** → EMPLOYEE

SOCIAL_INSURANCE (**Insurance_ID**, Employee_ID, Insurance_Number, Coverage_Details, Start_Date, End_Date, Status)

- References:
 - Employee_ID → EMPLOYEE.Employee_ID

EDUCATIONAL_QUALIFICATION (**Qualification_ID**, Employee_ID, Institution_Name, Major, Degree_Type)

- References:
 - Employee_ID → EMPLOYEE.Employee_ID

PROFESSIONAL_CERTIFICATE (**Certificate_ID**, Employee_ID, Certification_Name, Issuing_Organization, Issue_Date, Expiry_Date, Credential_ID)

- References:
 - Employee_ID → EMPLOYEE.Employee_ID

CONTRACT (**Contract_ID**, Contract_Name, Type, Description, Default_Duration, Work_Modality, Eligibility_Criteria)

JOB (**Job_ID**, Job_Code, Job_Title, Job_Level, Job_Category, Job_Grade, Min_Salary, Max_Salary, Job_Description, Status, Department_ID, Reports_To)

- References:
 - Department_ID → DEPARTMENT.Department_ID
 - Reports_To → JOB.Job_ID (self-reference)

JOB_OBJECTIVE (**Objective_ID**, Job_ID, Objective_Title, Description, Weight, Salary)

- References:
 - Job_ID → JOB.Job_ID

OBJECTIVE_KPI (**KPI_ID**, Objective_ID, KPI_Name, Description, Measurement_Unit, Target_Value, Weight)

- References:
 - Objective_ID → JOB_OBJECTIVE.Objective_ID

JOB_ASSIGNMENT (**Assignment_ID**, Employee_ID, Job_ID, Contract_ID, Start_Date, End_Date, Status, Assigned_Salary)

- References:
 - Employee_ID → EMPLOYEE.Employee_ID
 - Job_ID → JOB.Job_ID
 - Contract_ID → CONTRACT.Contract_ID

PERFORMANCE_CYCLE (**Cycle_ID**, Cycle_Name, Cycle_Type, Start_Date, End_Date, Submission_Deadline)

EMPLOYEE_KPI_SCORE (**Score_ID**, Assignment_ID, KPI_ID, Performance_Cycle_ID, Actual_Value, Employee_Score, Weighted_Score, Reviewer_ID, Comments, Review_Date)

- References:
 - Assignment_ID → JOB_ASSIGNMENT.Assignment_ID
 - KPI_ID → OBJECTIVE_KPI.KPI_ID
 - Performance_Cycle_ID → PERFORMANCE_CYCLE.Cycle_ID
 - Reviewer_ID → EMPLOYEE.Employee_ID

APPRAISAL (**Appraisal_ID**, Assignment_ID, Cycle_ID , Appraisal_Date, Overall_Score, Manager_Comments, HR_Comments, Employee_Comments, Reviewer_ID)

- References:
 - Assignment_ID → JOB_ASSIGNMENT.Assignment_ID
 - Cycle_ID → PERFORMANCE_CYCLE.Cycle_ID
 - Reviewer_ID → EMPLOYEE.Employee_ID

APPEAL (**Appeal_ID**, Appraisal_ID , Submission_Date, Reason, Original_Score, Approval_Status, appeal_outcome_Score)

- References:
 - Appraisal_ID → APPRAISAL.Appraisal_ID

TRAINING_PROGRAM (**Program_ID**, Program_Code, Title, Objectives, Type, Subtype, Delivery_Method, Approval_Status)

EMPLOYEE-TRAINING (**ET_ID** , Employee_ID , Program_ID , Completion_Status)

- References:
 - Employee_ID → EMPLOYEE.Employee_ID
 - Program_ID → TRAINING_PROGRAM.Program_ID

TRAINING_CERTIFICATE (**Certificate_ID**, **ET_ID** , Issue_Date, certificate_file_path)

- References:
 - ET_ID → EMPLOYEE_TRAINING.ET_ID